

GlowAI

SKIN CONDITION CLASSIFICATION FROM SYNTHETIC DATA

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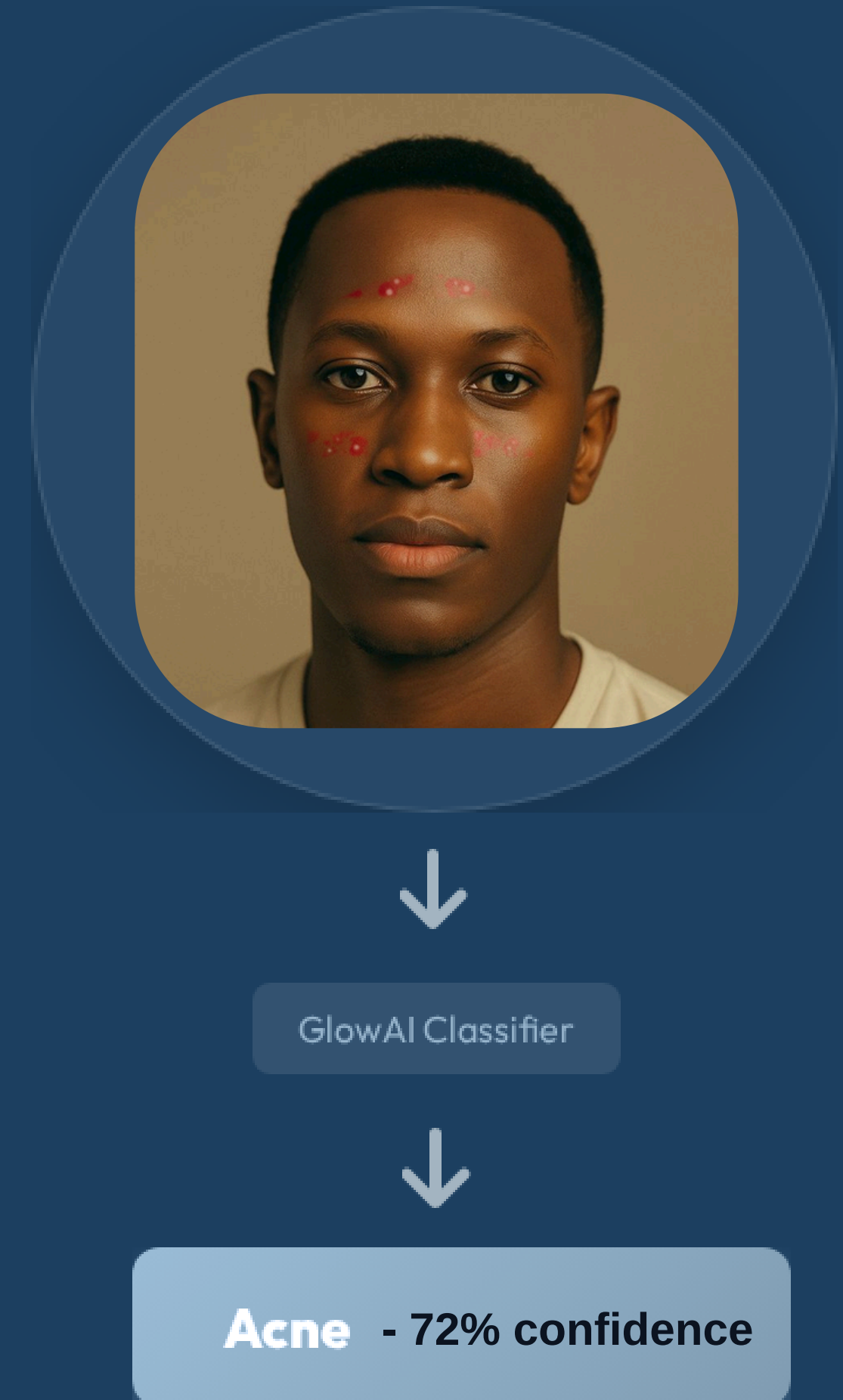
Project Definition

| Goal & Motivation

- ✓ **Goal:** Classify face images into three classes – **acne**, **erythema**, **none**.
- ✓ **Motivation:** Quick pre-screening from consumer-grade images (selfies), not a medical diagnosis.

| Input & Output

- ✓ **Input:** RGB face image (frontal selfie).
- ✓ **Output:** Predicted class + confidence score.



Project Achievements & Novelty

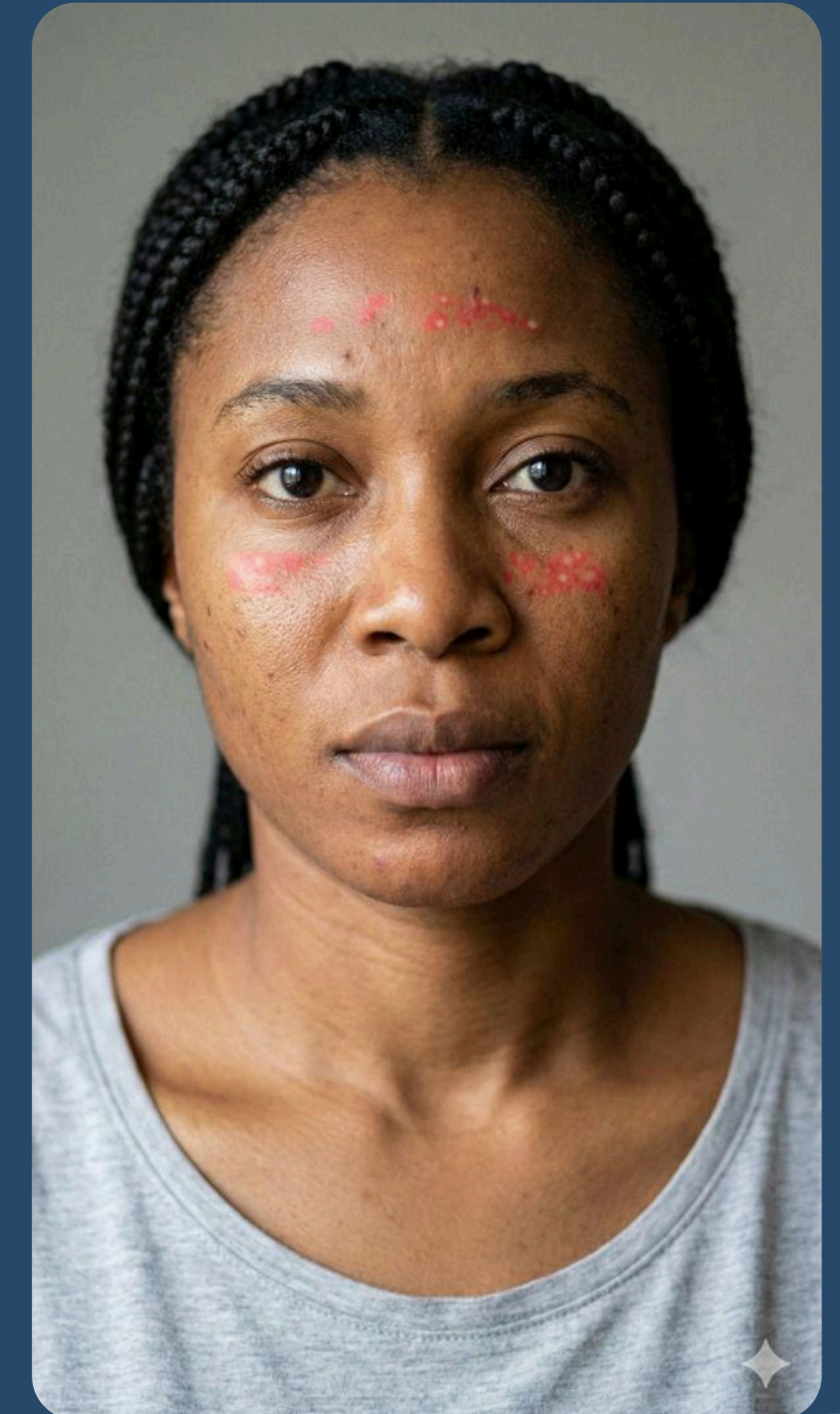
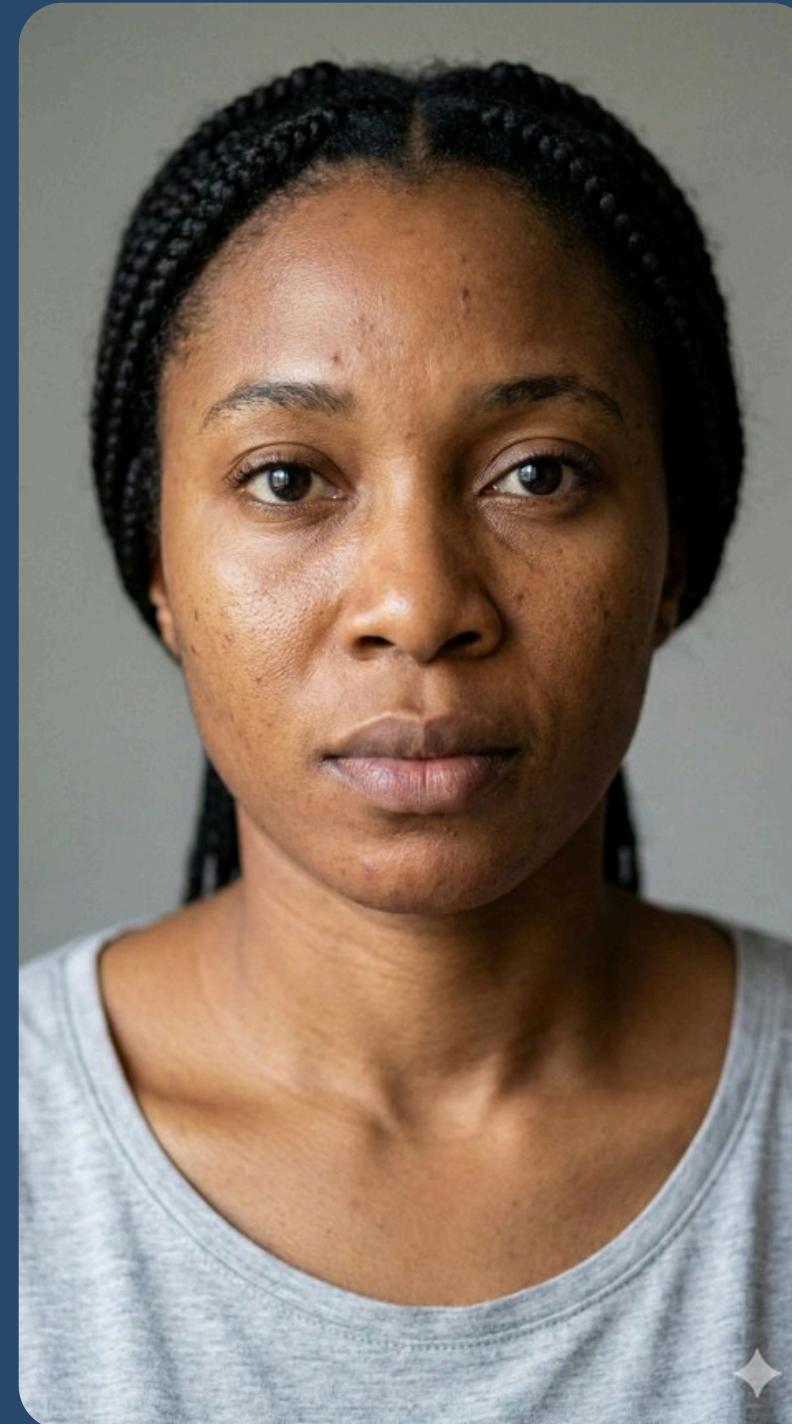
Achievements

- ✓ Trained a 3-class classifier (EfficientNet-B2) combining real and synthetic data.
- ✓ Built an end-to-end pipeline: generation → training → evaluation → feedback.

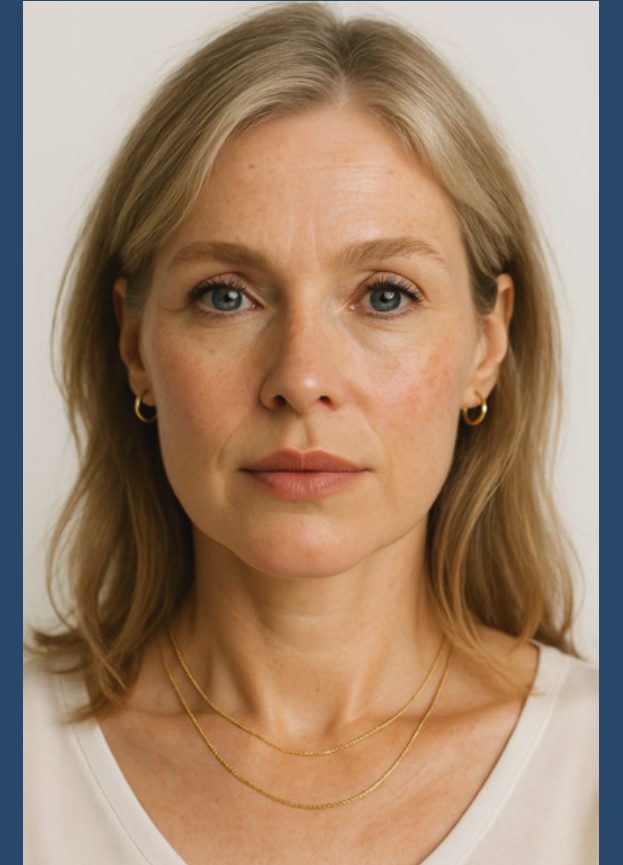
Novelty

- ✓ **Custom Skin Simulator:** OpenCV-based tool for realistic acne & erythema control (location, spread, intensity).
- ✓ **Synthetic Data Engine:** Solving scarcity of labeled medical data.
- ✓ **Feedback Loop:** Manual set to guide improvements.

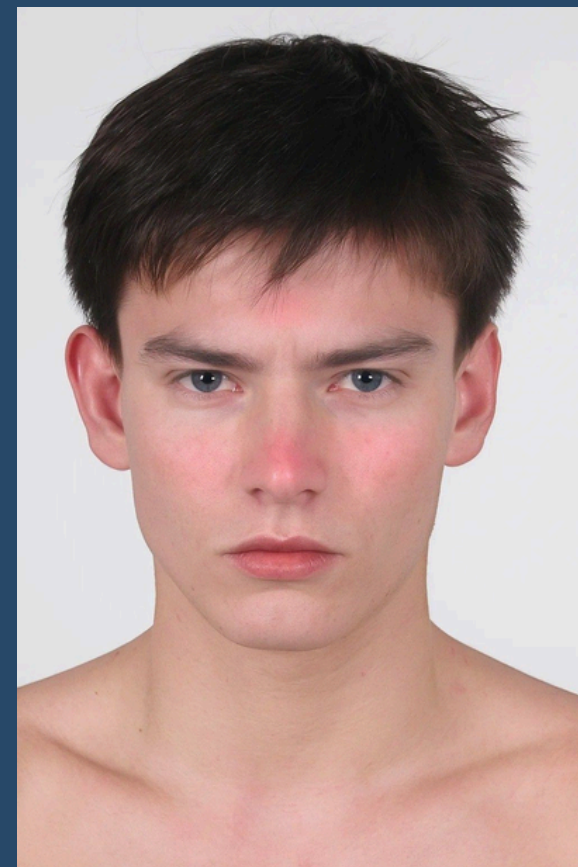
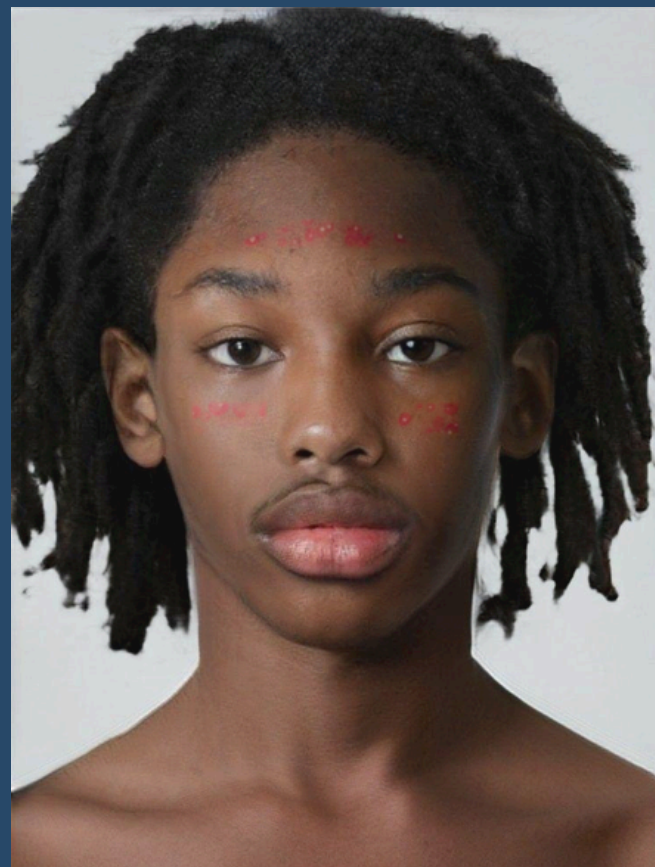
Comparison: Clean vs. Synthetic



BEFORE



AFTER



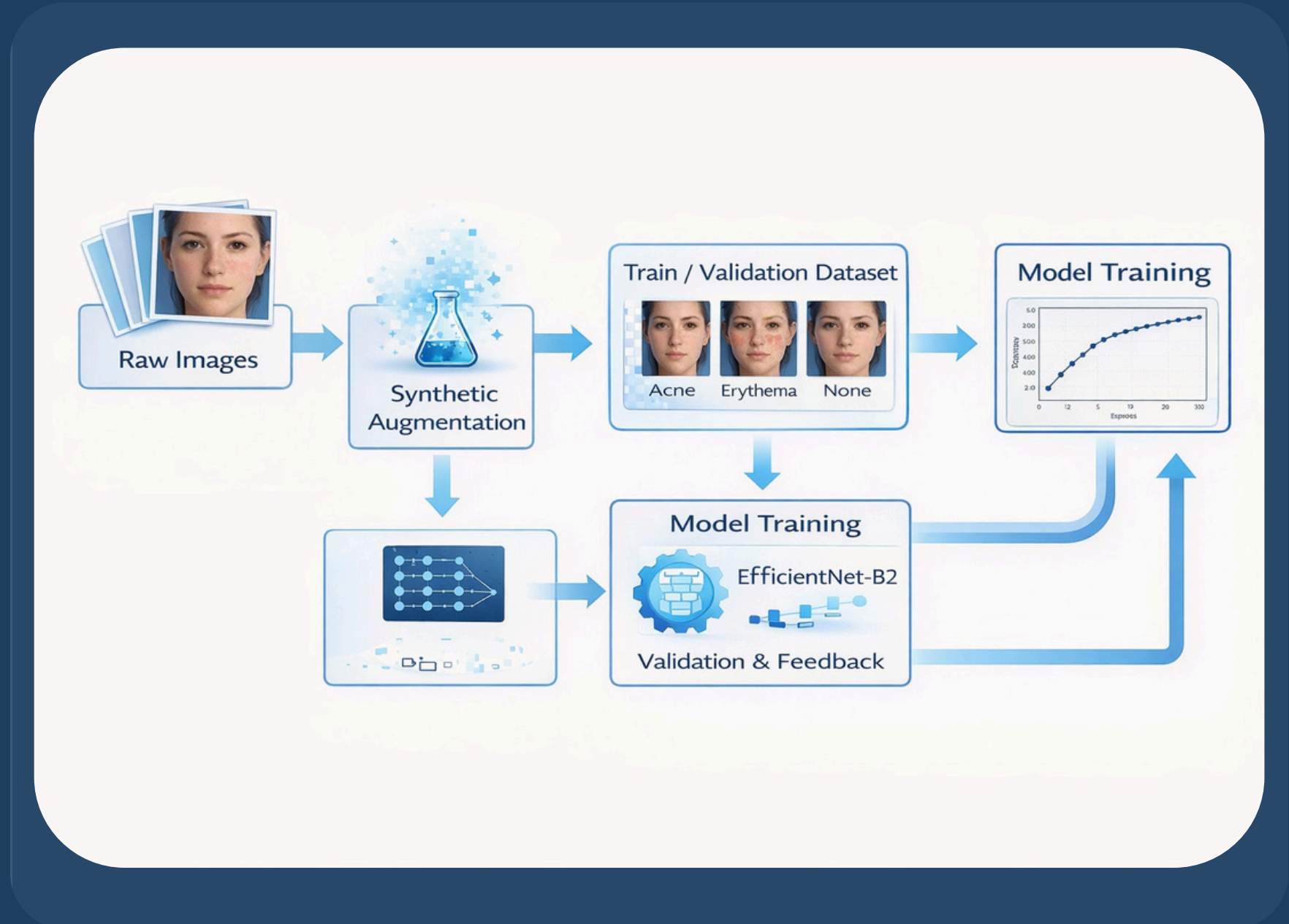
Methodology & Pipeline

Methodology

- ✓ **Data:** 3-class dataset (Acne/Erythema/None) mixed with synthetic data.
- ✓ **Preprocessing:** Resize 224×224, Normalize (ImageNet).
- ✓ **Augmentations:** Flip, rotation, color jitter (mimic selfies).
- ✓ **Training:** EfficientNet-B2, Two-stage fine-tuning, AdamW, Cross-Entropy.

Evaluation

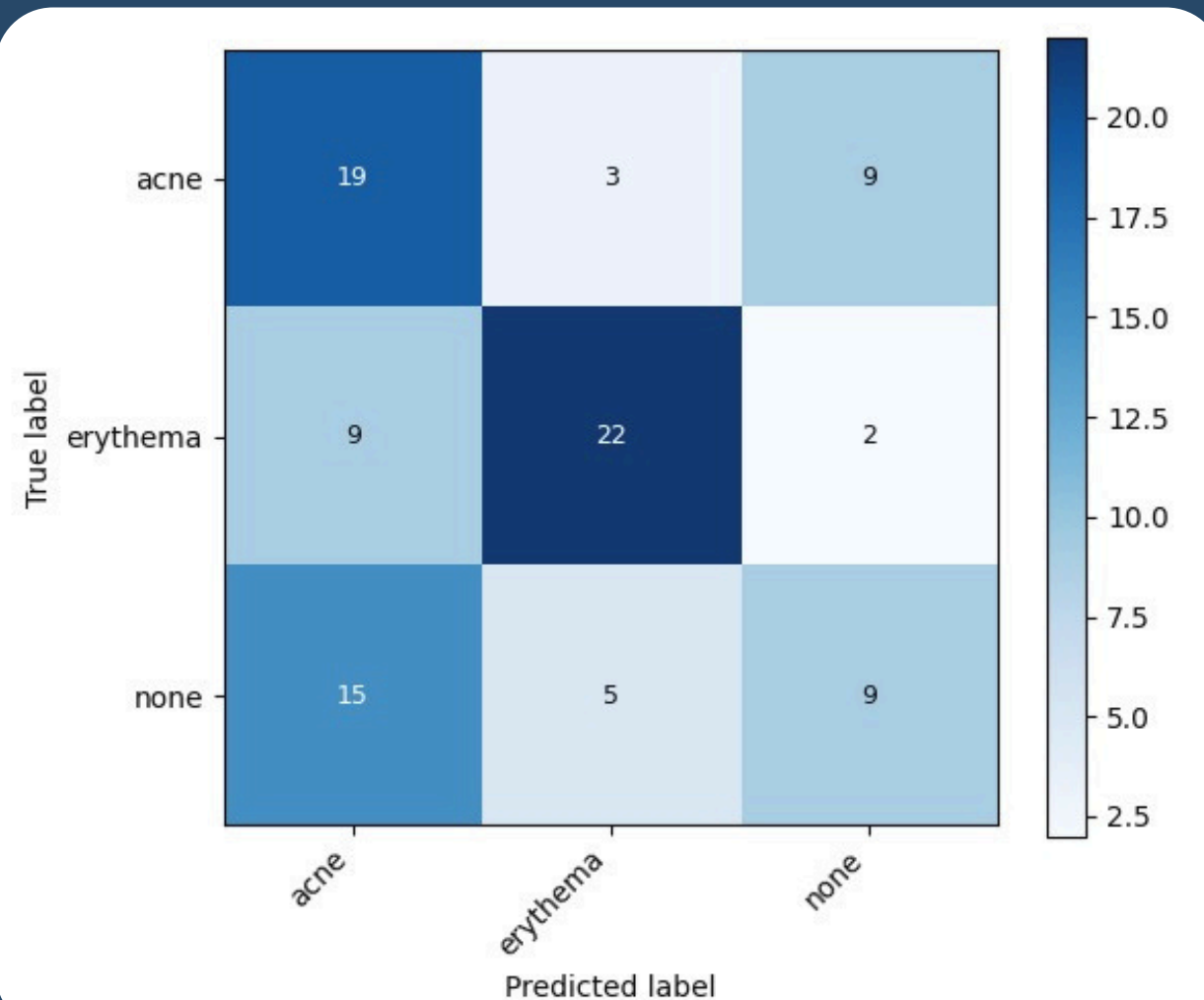
- ✓ Selected best model by validation accuracy.
- ✓ Analyzed per-class behavior via confusion matrix.



Results: Validation & Manual Test

Validation Results

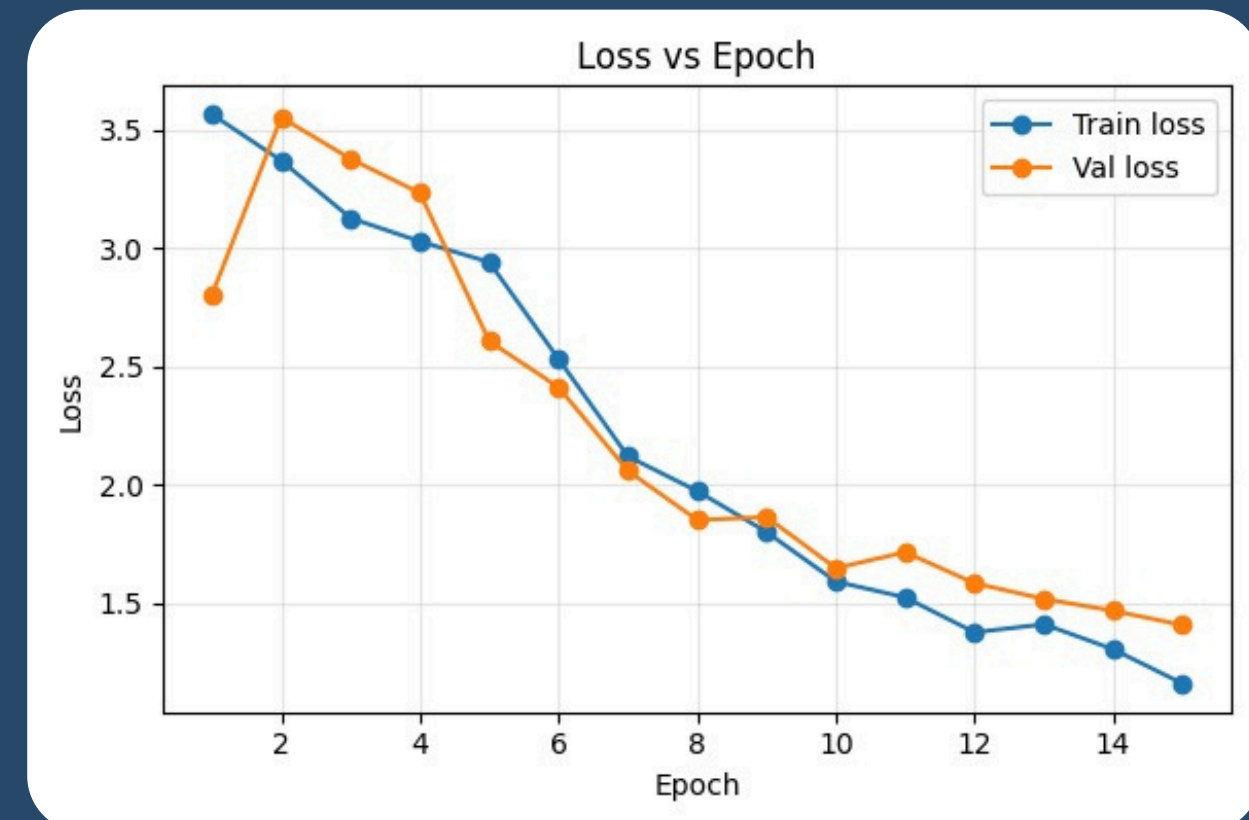
- ✓ Best Val Accuracy: ~54%.
- ✓ Main issue: "None" often predicted as "Acne/Erythema".



Confusion Matrix

Manual Test (Real Images)

- ✓ Acne: 11/15 | Erythema: 8/9 | None: 17/17
- ✓ Overall Accuracy: ~88% on curated real-world set.
- ✓ Significantly better performance on clean input than raw validation metrics suggest.



Training vs Validation Accuracy

Conclusion & Future Work

Conclusion

- ✓ Built full synthetic+real classifier pipeline.
- ✓ Reached ~**54% Val** / ~**88% Real Test** accuracy.
- ✓ Synthetic data is useful but highlights class overlap challenges.

Limitations

- ✓ Separating mild conditions is difficult.
- ✓ Class imbalance impacts robustness.
- ✓ Single-label setup is a simplification of reality.

Future Work

- ✓ Expand expert-labeled datasets.
- ✓ Move to pixel-level segmentation.
- ✓ Use diffusion models for better realism.

"These improvements could lead to a more accurate and clinically relevant skin analysis system."